

Matrix inequalities and norms

Modulus-order determinant comparison with an arbitrary base power

Difficulty: challenging **Importance:** interesting to specialist

Status: Solved

Rating rationale: Arbitrary base powers and indefinite Hermitian factors make extension of the squared-base theorem challenging; the comparison has specialist importance for determinant inequalities.

Resolution — 2026-09-11

Negative result by Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent proof review: PASS.**

A rational positive definite A and invertible indefinite Hermitian B in dimension three, with $k = 6$ and $p = 8$, reverse the proposed determinant comparison. The exact right-minus-left gap is $21036678407451/156250000000000 > 0$. The known $k = 2$ theorem and the variant $B > 0$ are not contradicted.

The exact target is resolved. The original statement and source evidence are retained below; its former difficulty rating is historical.

Primary manuscript: [complete proof PDF](#), [standalone TeX](#), Theorem 1.1 and its proof; [authorship and scope](#). The [independent review](#) checks the full original argument and records its hash. The draft was AI-assisted; this is independent agent verification, not external human peer review or formal certification. [Submission record](#).

Problem statement

Does the inequality

$$\det(A^k + |AB|^p) \geq \det(A^k + |BA|^p)$$

hold for every integer $n \geq 1$, every positive definite $A \in \mathbb{C}^{n \times n}$, every invertible Hermitian $B \in \mathbb{C}^{n \times n}$, and every pair of real parameters $k, p \geq 0$? Here $|X| = (X^*X)^{1/2}$, all powers of positive definite matrices use spectral functional calculus, and $X^0 = I$.

Both determinants are positive real numbers. The same conjecture is stated for semidefinite A and Hermitian B in the source; invertible inputs avoid zeroth-power conventions, while the positive-exponent singular cases follow by continuous approximation. No commutation between A and B is assumed.

The two moduli have the same singular spectrum, but adding A^k tests their alignment with the eigenvectors of the base matrix. The question therefore concerns how spectral data and eigenvector geometry interact in determinant estimates for matrix functions.

References

1. M. M. Ghabries, *Contributions to Matrix Inequalities and Some Applications*, PhD thesis, University of Angers and Lebanese University (2022), §2.5 and final Open Problems, Problem 2, pp. 109–110. [Thesis uploaded by its author.](#)
2. M. M. Ghabries, H. Abbas and B. Mourad, *On some open questions concerning determinantal inequalities*, *Linear Algebra Appl.* 596 (2020), 169–183, resolution of the earlier $k = 2$ question. [Publisher.](#)
3. M. M. Ghabries, *A log-majorization inequality for normal matrices with applications to determinantal inequalities and geometric means*, arXiv:2607.21163v1 (2026), §3. [Full text.](#)

Status check (2026-09-10): the thesis states the full question after proving $k = 2$ for all $p \geq 0$, and further cases when both matrices are positive semidefinite. The July 2026 paper extends the known squared-base result to normal matrices, but does not claim arbitrary k here. Searches for the exact titles, “modulus determinant arbitrary power conjecture Ghabries”, and 2025/2026 found no complete resolution. The negative-power companion in the thesis is not counted separately.